Table 1: Classical computing versus quantum computing

Classical computing	Quantum computing
Electromagnetism governs electric currents — classical electrodynamics	Quantum mechanics governs sub-atomic particles — quantum electrodynamics
Works on Moore's Law	Harnesses the power of sub-atoms and molecules to perform memory and processing tasks
Works with digital bits 0,1	Features quantum bits (qubits), superposition of 0 and 1
Deterministic	Stochastic
n-bits: 2^n sequences available, but only one used at a time	n-qbits: 2^n sequences available in parallel
Sequential computing	Parallel computing
One operation at a time	Millions of operations at a time
64-bit classical computer operates at speeds measured in gigaflops (billions of floating-point operations per second)	30-qubit quantum computer equals the processing power of conventional computers that run at 10 teraflops (trillions of floating-point operations per second)

concluded with the establishment of a new cloud-based computing service for trapped-ion quantum computers in Europe. The service grants European researchers access to devices from Alpine Quantum Technologies (AQT) to perform advanced quantum computing tasks, for innovation in healthcare and industrial advancements.

Google, Microsoft, SciQuantum, and IBM have achieved significant milestones in quantum computing, including new quantum processors, enhanced error correction techniques, and scalable quantum systems.

Quantum computers can be classified both by the physical qubit technology they employ and by the computational paradigm they implement (Table 2). Each approach comes with its own trade-offs in coherence time, gate fidelity, scalability, and operating conditions.

Characteristics of quantum computing platforms (QCPs)

Low level programming: Today's quantum computers are built on low level programming. Quantum logical gates handle the computational steps that are executed in quantum processing units (QPUs). Examples of quantum logical gates are Pauli, Hadamard, CCNot, etc.

Remote software development and deployment: The quantum computing software development framework for leveraging quantum processors is accessed remotely on the cloud. Only a limited portion of the programming tool stack is deployed on local machines.

Table 2: Types of quantum computers in use today

Quantum computer	Technology	Vendor/User
Superconductor-based quantum computers	Utilise superconducting circuits, including SQUID-based technology.	IBM's Eagle, Osprey, and Condor; Google's Sycamore and Quantum AI Horizon; Rigetti's Aspen-M
Ion trap-based quantum computers	Ion trap technology to achieve quantum computation.	IonQ (Aria), Quantinuum (H1-2), Alpine Quantum Technologies (PINE)
Neutral atom quantum computers	Use individual neutral atoms as qubits.	QuEra (Aquila and Aspen-M systems), ColdQuanta (Frostridge), Pasqal
Photonic quantum computers	Use photons as the fundamental qubit unit.	Xanadu and PsiQuantum
Surface quantum-dot quantum computers	Quantum dots formed in semiconductor material.	Intel and QuTech (Delft University), Silicon Quantum Computing (Australia), Quantum Motion Technologies (UK), HRL Laboratories, NIST, TU Delft, Microsoft StationQ (R&D)
Laser-cooled ion trap quantum computers	Use laser-cooled ions for quantum operations.	IonQ (Aria), Quantinuum (H1-2), Quantum Systems Accelerator (QSA), Alpine Quantum Technologies (PINE), GTRI QCCD
Solid state nuclear magnetic resonance (NMR) Kane quantum computers	Use solid state NMR techniques.	Bruce Kane (UNSW), NIST/JQI, STM-Lithography
Adiabatic quantum computers	Quantum annealing for computation	D-Wave